PREDICTION OF VICARIOUS TRIAL AND ERROR BY MEANS OF THE SCHEMATIC SOWBUG [1]

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The present report consists of two parts: first, the presentation of some experimental results; and second, the presentation of a theoretical schema for the interpretation of those results.

The Experiment

The experiment was a white-black and a white-gray discrimination experiment but the interest was not simply in the relative ease with which the subjects (rats) learned to differentiate between the correct and incorrect stimulus-objects but also in the relative amounts of 'looking back and forth' or, to use Muenzinger's term, 'vicarious trial and error' (or 'VTE') which they exhibited in the course of acquiring these differentiations.[2] In short, our experiment will be concerned not only with learning scores but also with VTE scores.

Three groups of rats with pigmented eyes (all approximately five months old at the beginning of the experiment) were run in a modified Lashley discrimination set-up. See Fig. 1. For all these groups the correct door was white, but for one the incorrect door was black, for another it was medium gray, and for the third it was a very light gray. The group run on the white-black set-up contained 9 males and 2 females, that on the white-medium gray set-up 8 males and 2 females, and that on the white-very light gray set-up 8 males and 2 females. The males were all run at the beginning and the

[1] Read, in part, before the American Psychological Association, September 7, 1938.

[2] For a discussion and summary of most of what is known to date about vicarious trial and error, much of it obtained in the Colorado laboratory, see Muenzinger's recent splendid report (7). See also Tolman (9; 10, Chap. XIII; 11) and Dennis & Russell (5).

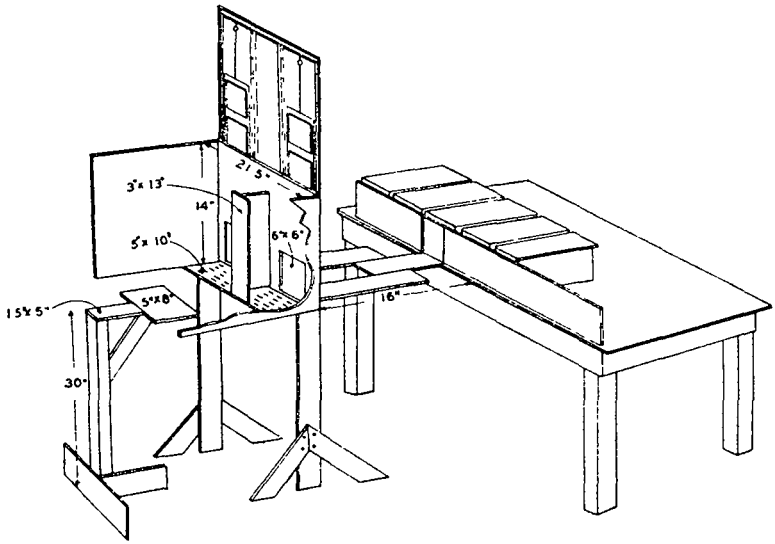

3"x 13"
14"
5"x 10"
2½ 5"
6"x 6"
15½ 5"
5"x 8"
16"
30"

FIG. 1

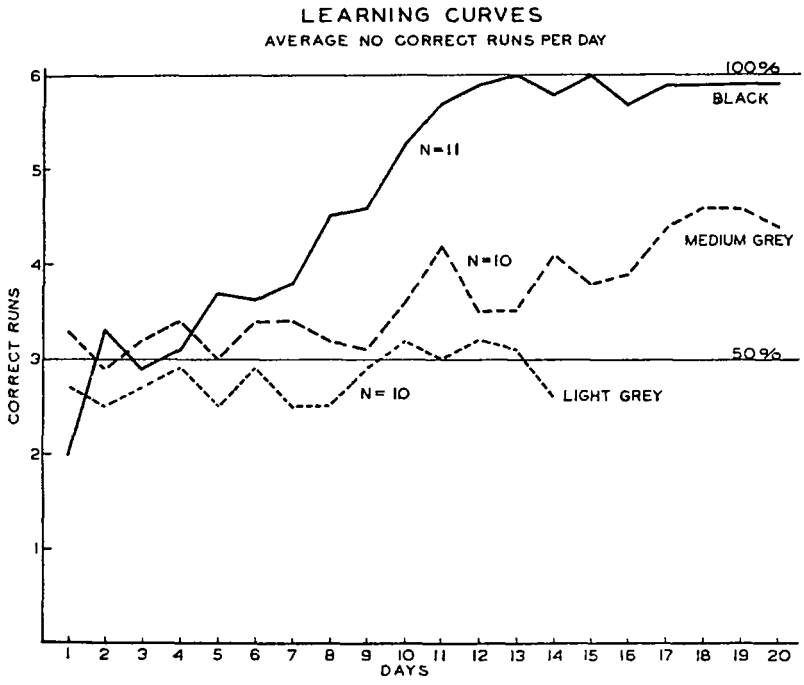

LEARNING CURVES
AVERAGE NO CORRECT RUNS PER DAY
100%
BLACK
N=11
N=10
MEDIUM GREY
50%
N= 10
LIGHT GREY
CORRECT RUNS
DAYS
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

FIG. 2

females all run at the end of each day's session. A preliminary training period consisting of seven experimental days in which the animals were taught to jump and to push open the doors preceded the experiment proper. In the preliminary period both doors were white. In the experiment proper every rat was given six trials a day. The positions of the correct door were on alternate days, LRRLLR and RLLRRL for all three groups. The gap during the experiment proper was eight inches wide for the males and six inches wide for the females. The animals were not run on Saturdays or Sundays. They were given a heavy meal in their cages on Saturdays and were not fed at all Sundays. No evidence of any disturbance in the learning curves appeared as a result of this procedure.

The learning curves for the three groups are shown in Fig. 2. The white-black group learned best, the white-medium gray next best, and the white-very light gray not at all.

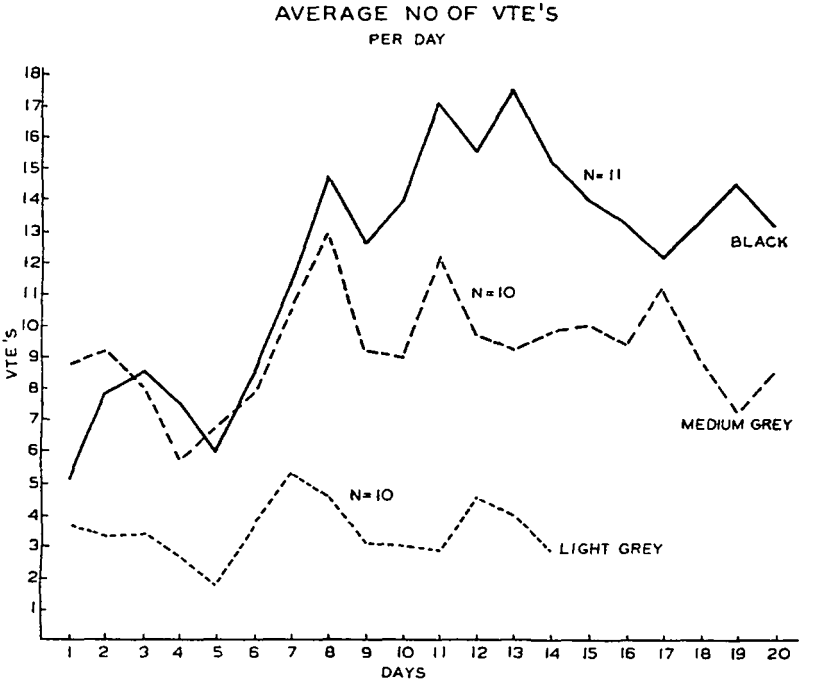


FIG. 3

Each look by a rat at one door, which was followed by a look (or a final taking-off-toward) the other door, was counted as one 'vicarious trial and error'—one VTE. For each group the average VTE's per rat in the six trials of each day are shown in Fig. 3. The white-black group exhibited the most. the white-medium gray group the next most, and the white-very-light-gray group the fewest VTE's.

Figure 4 presents the time curves. These are the times

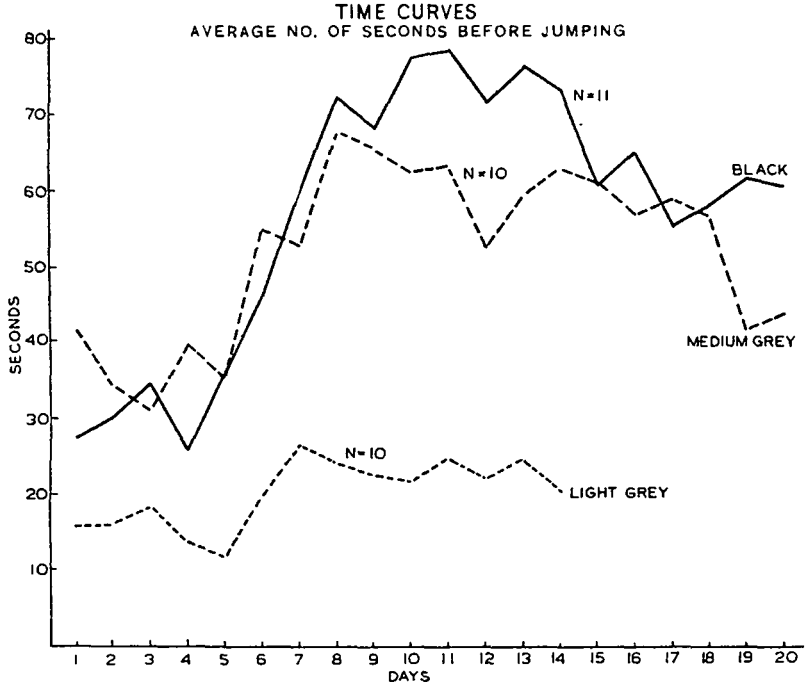


FIG. 4

from the instant the animal was put on the stand until he first took off either to the correct or to the incorrect door. It will be observed that these time curves are quite similar to the VTE curves.

Finally, in order further to emphasize the picture, curves for a couple of single individuals are presented.

Figure 5 shows the curves for an individual in the white-black group. This animal exhibited a complete and unvary-

ing left-going position habit for the first seven days. He averaged only a second or two before taking off to the left, irrespective of whether this was correct or incorrect. On the third trial of the 8th day, however, he slowed up, made one VTE, took 16 seconds, and then jumped to the right-hand side, and this was correct. From then on he began making more

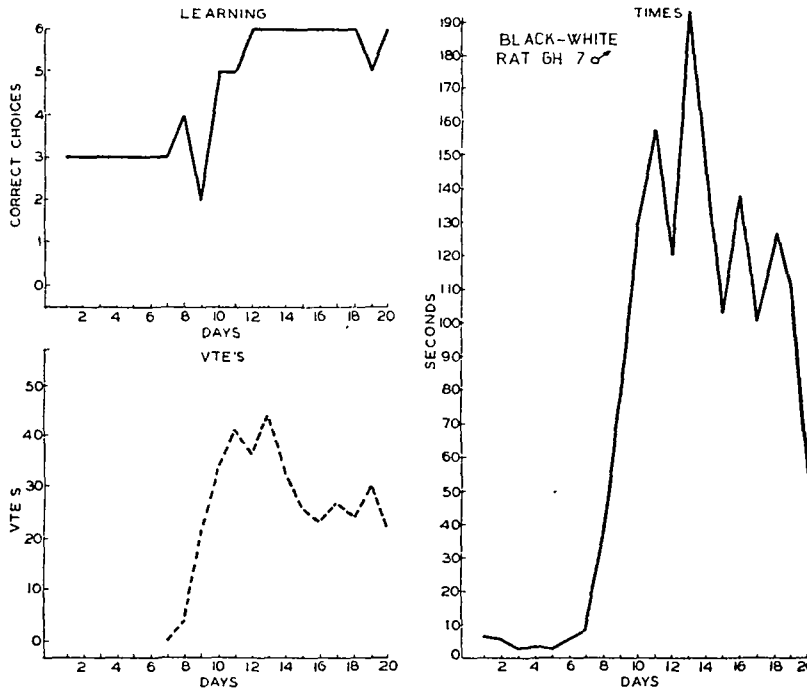


FIG. 5

and more VTE's and averaging more correct jumps even when they were to his previously avoided right-hand side.

Figure 6 shows the same three curves, for an individual in the white-medium-gray group. It will be observed that the learning did not become so complete and the VTE's and the longer times did not begin so soon and did not begin to come down again with any such certainty as was the case for the rat in the white-black group.[3]

[3] These two animals were selected since they were deemed 'typical.' It must be admitted, however, that there was considerable overlapping between the two groups.

So much for the facts; let us turn now to the theory.

The Theory

The theory (which in its present stage should not be taken too seriously) consists in the assumption that all higher organisms are in a certain *underlying* sense tropistic. This, not literally but figuratively speaking, tropistic or taxic character

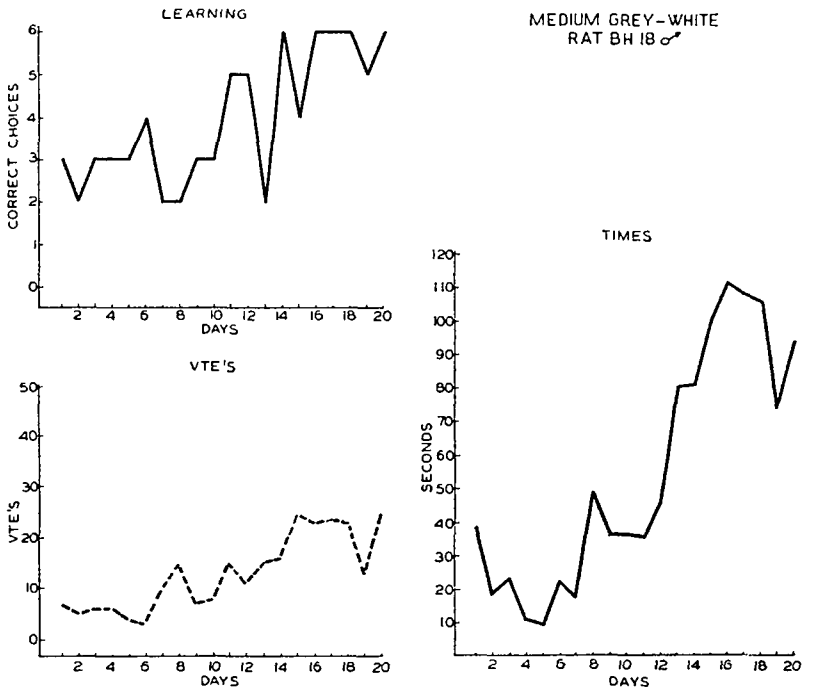


Fig. 6

of higher organisms provides, I believe, the necessary model for stating how in this experiment with rats different strengths of hunger, different goodnesses of perceptual differentiations, and different degrees of learning, combine to produce the actual amounts of correct choosing and of VTE-ing that they do. Or, to put it another way (see Fig. 7), the schematic

In particular, it is to be noted that there were one or two rats in the white-black group who never made many VTE's and yet learned quickly. Their VTE's, what there were of them, did, however, go up just before learning and then come down again.

sowbug is my present conception of what I have elsewhere (11) merely labelled as the f_3-function. The name 'schematic sowbug' is, of course, merely a would-be picturesque term for this essentially tropistic character which I have introduced into my f_3-function.

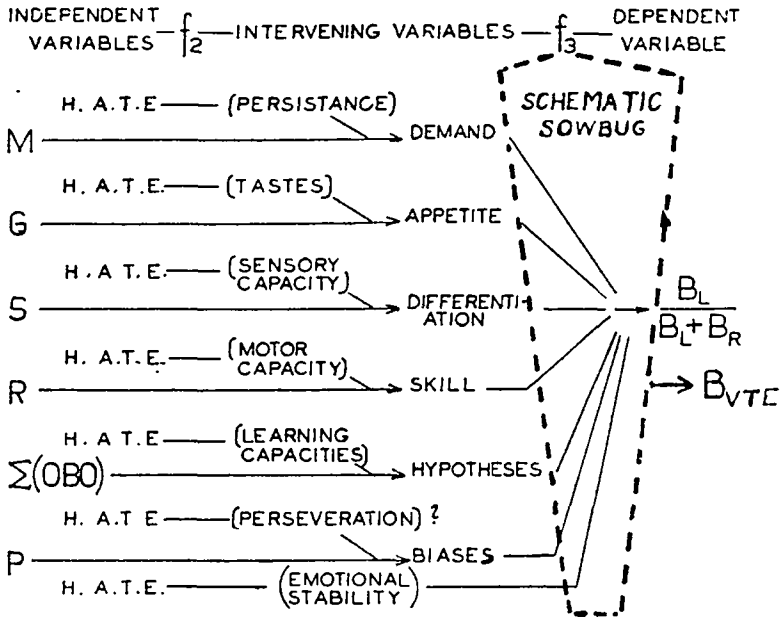


FIG. 7. The 'Schematic Sowbug' as the f_3-function.

Blum (1, 2, 3, 4), working with simple actually tropistic animals, discovered that many of the apparent deviations from the forced-movement type of behavior to be predicted from Loeb's theory could be explained by assuming that progression velocity (forward or backward) may vary independently of orientation velocity. I have taken over this distinction between orientation and progression for my bug and then I have assumed, further, that any set of qualities, which form a discriminable dimension for the given species, can be represented conceptually as lying at different angles around the nose of such a bug. Thus, if two qualities in such a dimension are relatively near together, they will be represented as lying in the sowbug's schematic space with a narrow angle between

them; but, if the two qualities are discriminably far apart, they will be represented as having a wider angle between them. The white and the black in the present experiment I would represent, for example, as lying some 90 degrees apart in the sowbug's space, but the white and the medium gray as lying, say, only some 60 degrees apart in this same space.

Figure 8 presents a first diagram of the bug. It is shown

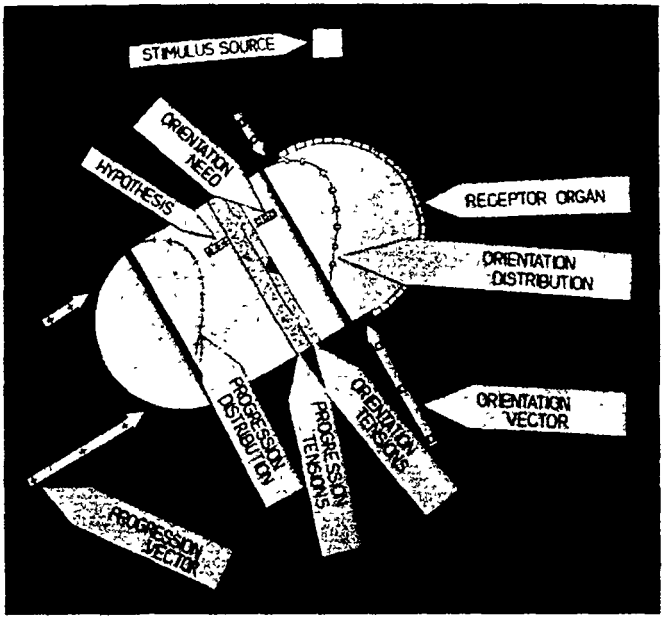


FIG. 8. The 'Schematic Sowbug.'

as lying at an angle to the parallel rays from a single stimulus source, say, white. It is bilaterally symmetrical. It has symmetrically placed pairs of motor appendages. These latter, however, are represented by vector arrows rather than as actual motor organs. The pair with small white rectangles on them, at the front just behind the head, are the orientation vectors, the pair with the plus marks on them at the rear are the progression vectors.

The curve with small rectangles labelled 'orientation distribution' indicates the relative intensities of perceptual

stimulation of the successive receptor points. These strengths of excitation are to be conceived as determined by the angles at which the rays hit plus the angular distances of the receptor points in question from the animal's median plane. The more perpendicularly a ray hits and the nearer the receptor point, which it hits, is to the nose (*i.e.*, the median plane), the stronger the excitation. When all the rays come (as in this case) from the left, the distribution is skewed and its mode lies to the left of the median plane. But when the rays come straight on, the orientation distribution becomes symmetrical. It is also to be assumed, following Loeb, that the right-hand motor appendages are activated by the left-hand receptors, and the left-hand motor appendages by the right-hand receptors. (I am assuming, that is, a positively taxic bug.) In other words, the area under the orientation distribution which is to the left of the median plane determines the length of the right-hand orientation vector, and the area under this curve which is to the right of the median plane determines the length of the left-hand orientation vector.

Further, it must be noted that although the shape of the orientation distribution is determined by the angular direction of the rays, its height is determined, rather, by a specific 'orientation need' relative to the given quality. This specific need is indicated by the little column with rectangles in it which is shown as rising up out of the upper stippled area, which latter represents the general orientation readiness or tension. This way of representing the tension or readiness as a stippled area inside the organism, the degree of stippling corresponding to the degree of the demand (derived in this case from hunger), I have adapted from Lewin (**6**). Further, it is also to be assumed that once the bug has looked directly at a given stimulus source the specific need relative to this specific quality and the corresponding resultant orientation distribution both temporarily sink.

The rear curve with the plus signs—the progression distribution—is to be conceived as following as to shape the orientation distribution. That is to say, its mode will shift following that of the orientation distribution when and if the

animal turns. The height of this progression distribution is to be conceived, however, as determined by the strength of the specific hypothesis—represented by the lower column with the plus signs on it—which hypothesis, in this case, is to the effect that the given stimulus source is good. This hypothesis in its turn is a product of the general progression tension—represented by the lower stippled area—plus specific past experience relative to this specific type of stimulus

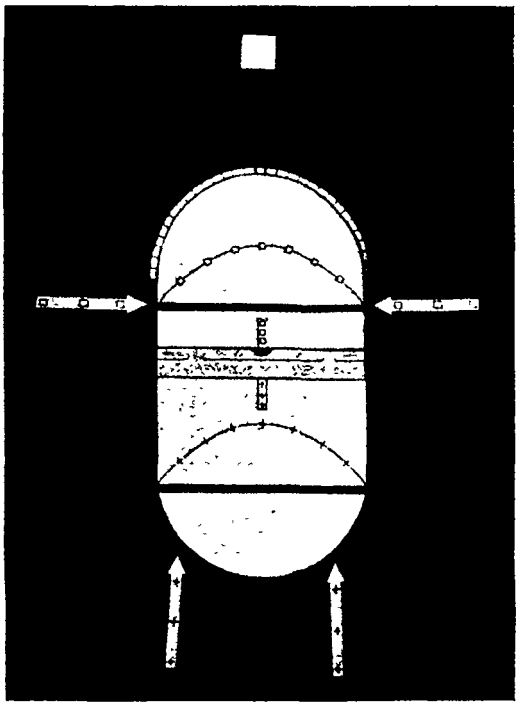


FIG. 9

source. It is such past experience which makes such a plus hypothesis ready to erupt. Note, further, that it is the right-hand and left-hand areas under the progression distribution which determine the relative lengths of the two progression vectors.

It is obvious that as here indicated the bug is in unstable equilibrium and will turn left until, as shown in Fig. 9, it will

face the stimulus, where its orientation and progression distributions will be symmetrical and, if nothing else happens, it will proceed to progress forward directly towards the stimulus source.

Now let us turn to the case of two competing stimuli, for this is the situation represented in our experiments, and the one in which VTE-ing occurs. The next two figures show the

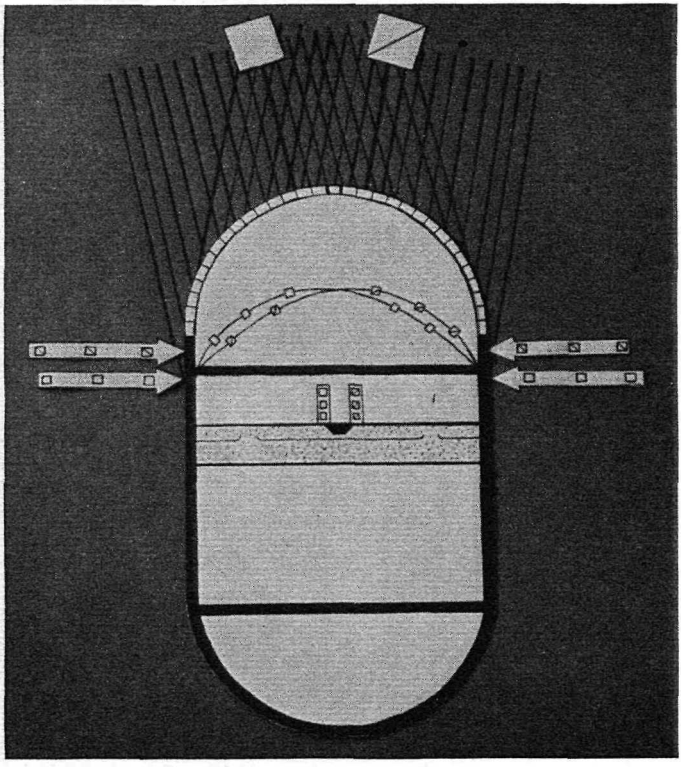


FIG. 10. Difficult discrimination.

relative sizes of the orientation vectors, first for a difficult discrimination (Fig. 10) between the two stimuli (represented in the figure by a white rectangle and a white rectangle with a diagonal line) and second for an easy discrimination (Fig. 11) between the two stimuli (represented in the figure by a white rectangle and a black rectangle). It is obvious that in this

latter case the imbalance of the orientation vectors is relatively great. Both figures, however, show the animal in the doldrums. In both cases he is like Buridan's ass between the two equidistant bundles of hay. Suppose, however, that he has just looked directly at one of the two stimuli, then our assumption is that his orientation need for that one,

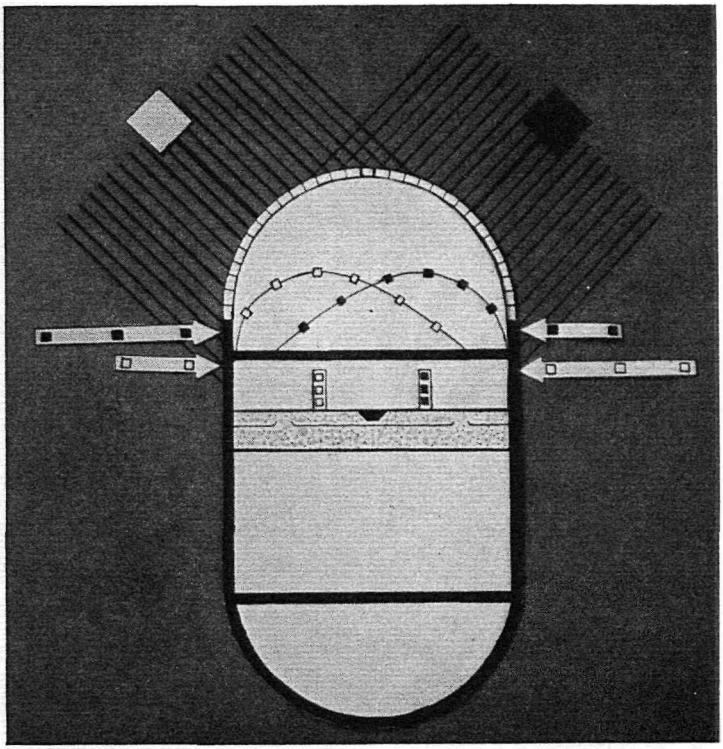


FIG. 11. Easy discrimination.

say white, will have become temporarily depressed. His white orientation distribution will have become lower and his white orientation vectors will have both become relatively short and he will swing over toward the right. But thereupon his orientation need with regard to the right-hand stimulus will, we assume, also become temporarily weaker while the one for the left-hand stimulus will recover and so he will swing back again. In short, he will VTE.

Why, however, does this tendency for VTE-ing appear sooner for the white and black choice than for the white and gray choices? As was said earlier, all the animals tended to start with initial position-habits—that is to say, initial irrelevant hypotheses. But an already established hypothesis will tend as such to interfere with VTE-ing. For, if this

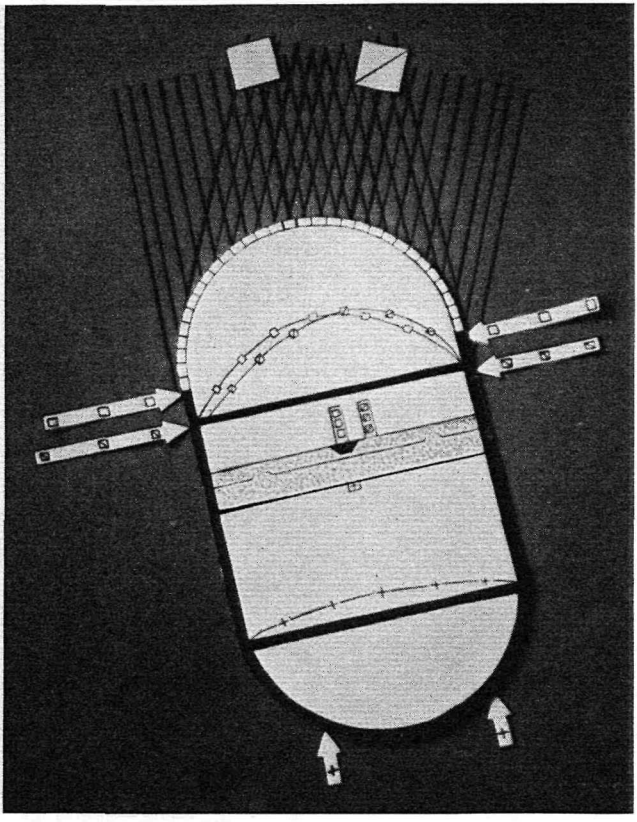


FIG. 12. Difficult discrimination.

hypothesis is strong enough, the animal will simply progress toward the given side and the imbalance in the orientation vectors relative to the brightness value of two qualities will not have a chance to get in its effects.

The next three figures indicate the relative heights of such already established initial hypotheses which would be big

enough thus to *prevent* VTE-ing. First, for a difficult discrimination (Fig. 12). It will be seen that here only a small initial irrelevant hypothesis in favor of the one side will be enough thus to prevent VTE-ing from occurring. Next consider a mediumly difficult discrimination (Fig. 13). Here

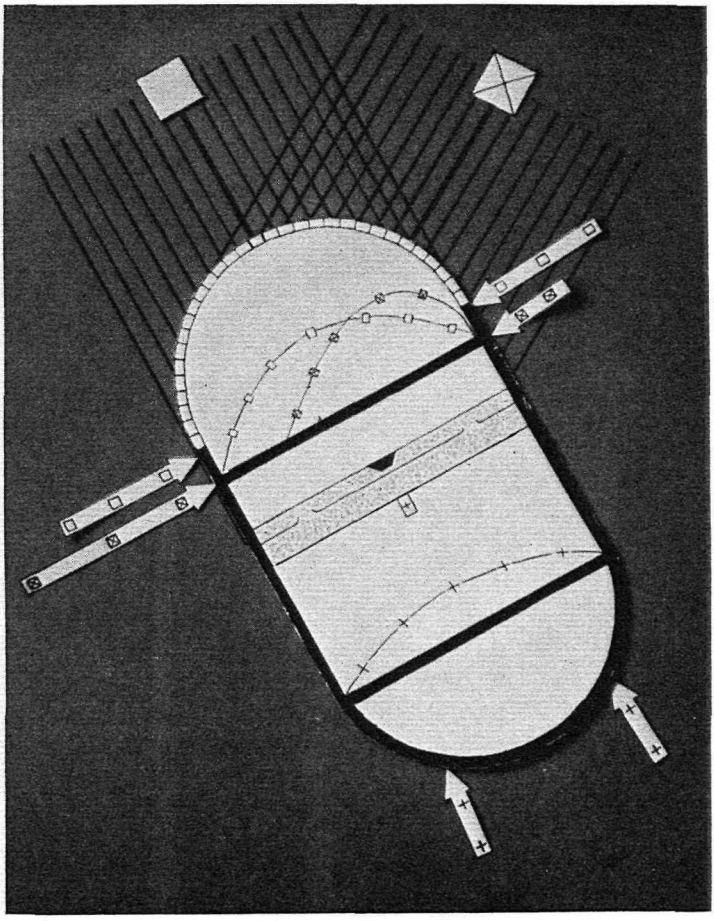


FIG. 13. Medium discrimination.

a stronger initial irrelevant hypothesis would be necessary to prevent VTE-ing. And, finally, for a still easier discrimination (Fig. 14) a still stronger initial hypothesis would be necessary to prevent VTE-ing. Hence VTE-ing starts more

readily for the white and black easy discrimination than for the more difficult white and medium gray discrimination and still more readily than for the white and very light gray very difficult discrimination. The initial irrelevant hypotheses are less likely to be strong enough to prevent VTE-ing in the case of the easier discriminations.

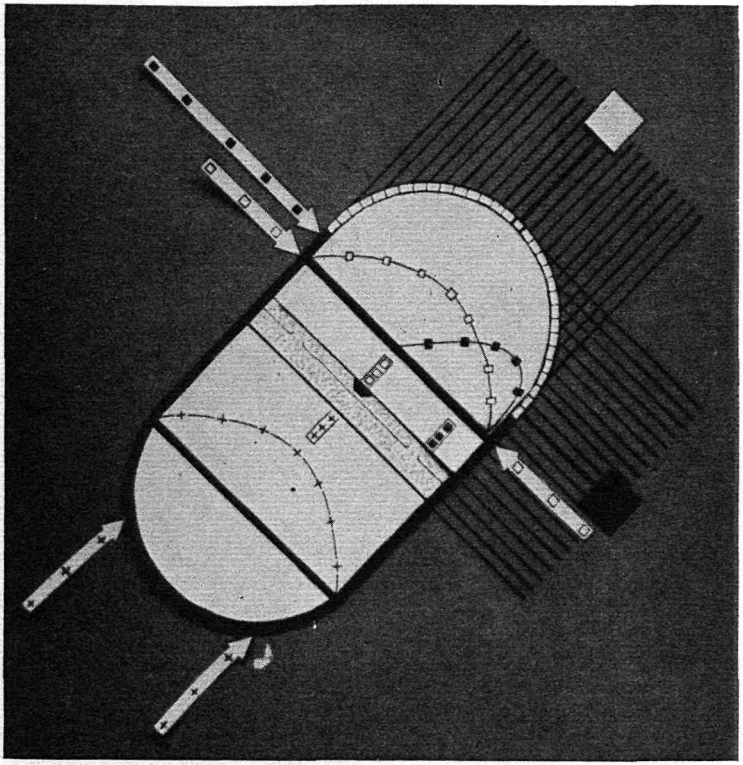


FIG. 14. Easy discrimination.

Finally, let us consider the course of learning. Figure 15 shows the situation, as I conceive it, early in learning for a white vs. medium gray discrimination. The initial irrelevant hypothesis has through lack of success been displaced and correct hypotheses have through the repetitions of success and failure begun to be set up. That is to say, a plus progression curve has started for the plus (white) stimulus source and

a minus progression has started for the minus (gray) stimulus source. There are plus forward-*pushing* vectors and minus backward-*pulling* vectors. These resultant progression vectors both plus and minus are, however, still relatively small, and VTE-ing still occurs.

But such VTE's, as soon as they appear, are, I shall assume, a help. I shall suppose, in short, that each time the

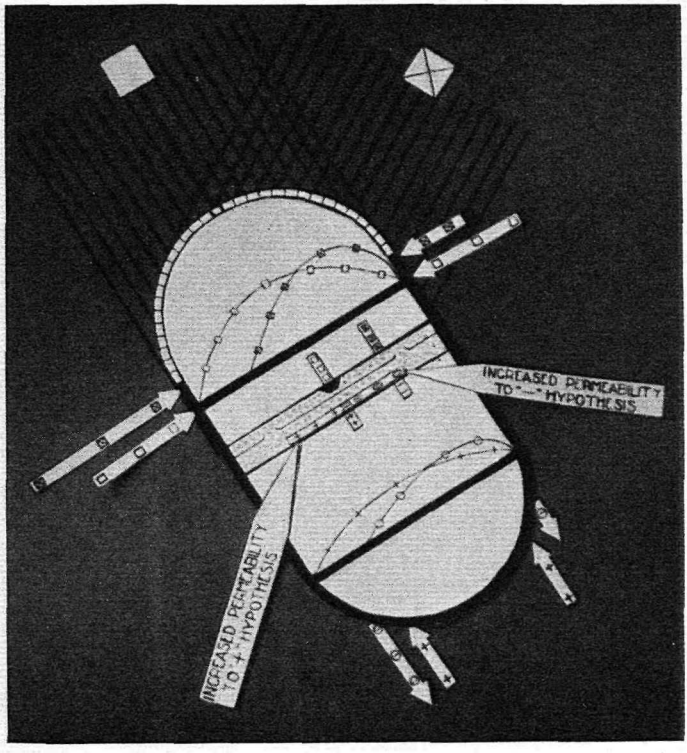


FIG. 15. Early in learning.

rat VTE's and looks at one or the other door there is some slight tendency for the effect of the correct hypotheses already forming to be reinforced. And since VTE-ing occurs more readily for the white-black group, this means that the white-black learning tends to be faster than this white-medium-gray learning. Figure 16 shows the situation later in learning.

Here the positive and negative vectors have become very strong. They have overcome the imbalance of the orientation vectors. There is practically no VTE-ing, and the bug now progresses directly towards the positive (*i.e.*, white) stimulus.

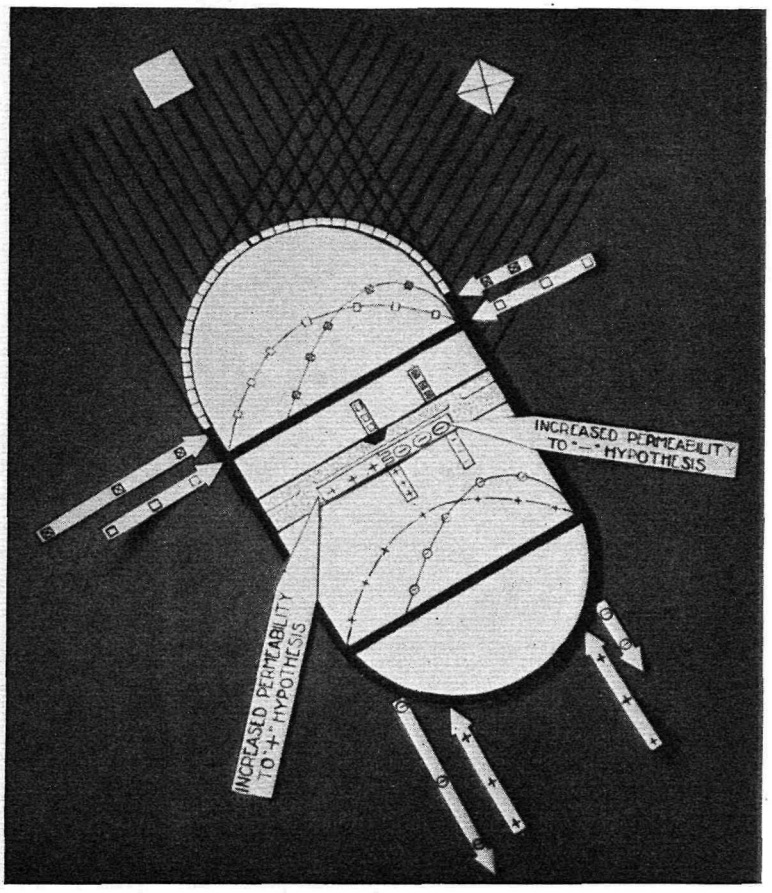


FIG. 16. Late in learning.

Another point about both Figs. 15 and 16 is very important. It will be noted that along the lower membrane of the progression tension I have tried to indicate the fact of increased permeabilities to the hypotheses which are being acquired. That is to say, as learning progresses the right-hand side of this membrane is conceived as becoming more and

more permeable to the negative hypothesis and the left-hand side of the membrane as becoming more and more permeable to the positive hypothesis. The effect of learning is thus conceptualized by increases in the respective permeabilities to the two hypotheses.[4]

One last word. This schematic sowbug, however bizarre and silly it may appear, is of course in no important sense original. All its more essential characteristics have been proposed by others. I conceive it to be patterned very much after Lewin's psychological 'person' as he draws such a person in a 'life space' (6). What I have done is to add to Lewin's diagram a nose and a tail and more specific receptor and motor mechanisms; I have also added the notion of spatially representing discriminable arrays of qualities as different angles in the 'life space.' And, of course, as I have already pointed out, it was from Loeb and Blum that I borrowed the notion of the manner of interconnection of these sensory and motor mechanisms according to the forced movement doctrine of tropisms. Again, it was from Spence (8)—though he may not like it— that the notion of the orientation and progression curves was taken. I have merely adapted Spence's generalization curves and put them inside my bug.

Finally, I wish to express my very great indebtedness to Mr. Clarke W. Crannell for his help in designing some of the actual details of the bug and for making the photographs from which the above figures have been copied.

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[4] Another and possibly better way to have conceived the bug might have been to have called the permeabilities the hypotheses and then to have called the resultant projecting columns not hypotheses, but progression needs. The height of the progression needs would then have been said to result from (*a*) the intensities of the general progression tension together with (*b*) the strengths of the corresponding hypotheses (*i.e.*, the specific degrees of permeability, respectively, to plus progressions and to minus progressions).

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